

**SAFETY DATA SHEET**

Revision Date 29-March-2024

Revision Number 4

**1. Identification**

**Product Name** 2-Methyl-5-nitroaniline

**Cat No. :** A17726

**CAS-No** 99-55-8  
**Synonyms** 2-Amino-4-nitrotoluene; C.I. 37105; 5-Nitro-o-toluidine

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company**

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**2. Hazard(s) identification****Classification**

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                           |            |
|---------------------------|------------|
| Acute oral toxicity       | Category 3 |
| Acute dermal toxicity     | Category 3 |
| Acute Inhalation Toxicity | Category 3 |
| Carcinogenicity           | Category 2 |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

Toxic if swallowed, in contact with skin or if inhaled  
Suspected of causing cancer

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Wear protective gloves/protective clothing/eye protection/face protection  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area

**Response**

IF exposed or concerned: Get medical advice/attention  
IF SWALLOWED: Immediately call a POISON CENTER/doctor  
IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Call a POISON CENTER/ doctor  
Rinse mouth  
Take off immediately all contaminated clothing and wash it before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Harmful to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component                      | CAS-No  | Weight % |
|--------------------------------|---------|----------|
| Benzenamine, 2-methyl-5-nitro- | 99-55-8 | 96       |

### 4. First-aid measures

|                                        |                                                                                                                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                    |
| <b>Skin Contact</b>                    | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.                                       |
| <b>Inhalation</b>                      | Remove from exposure, lie down. Remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Immediate medical attention is required. |
| <b>Ingestion</b>                       | Call a physician immediately. Clean mouth with water.                                                                                                                                |
| <b>Most important symptoms/effects</b> | No information available.                                                                                                                                                            |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                |

### 5. Fire-fighting measures

**Suitable Extinguishing Media** Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Alcohol resistant foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
3

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

See Section 12 for additional Ecological Information. Avoid release to the environment. Collect spillage. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal.

## 7. Handling and storage

**Handling**

Do not breathe dust. Do not get in eyes, on skin, or on clothing. Handle product only in closed system or provide appropriate exhaust ventilation.

**Storage.**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Incompatible Materials. Acids. Strong oxidizing agents. Acid anhydrides. Acid chlorides. Chloroformates.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component                      | Alberta | British Columbia         | Ontario TWAEV            | Quebec | ACGIH TLV                | OSHA PEL | NIOSH |
|--------------------------------|---------|--------------------------|--------------------------|--------|--------------------------|----------|-------|
| Benzenamine, 2-methyl-5-nitro- |         | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> |        | TWA: 1 mg/m <sup>3</sup> |          |       |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**  
**Hand Protection**Wear safety glasses with side shields (or goggles) Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

Wear a NIOSH/MSHA or European Standard EN 149 approved full-facepiece airline respirator in the positive pressure mode with emergency escape provisions.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**9. Physical and chemical properties**

|                                               |                                 |
|-----------------------------------------------|---------------------------------|
| <b>Physical State</b>                         | Powder Solid                    |
| <b>Appearance</b>                             | Amber                           |
| <b>Odor</b>                                   | No information available        |
| <b>Odor Threshold</b>                         | No information available        |
| <b>pH</b>                                     | No information available        |
| <b>Melting Point/Range</b>                    | 104 - 107 °C / 219.2 - 224.6 °F |
| <b>Boiling Point/Range</b>                    | No information available        |
| <b>Flash Point</b>                            | No information available        |
| <b>Evaporation Rate</b>                       | Not applicable                  |
| <b>Flammability (solid,gas)</b>               | No information available        |
| <b>Flammability or explosive limits</b>       |                                 |
| Upper                                         | No data available               |
| Lower                                         | No data available               |
| <b>Vapor Pressure</b>                         | No information available        |
| <b>Vapor Density</b>                          | Not applicable                  |
| <b>Specific Gravity</b>                       | No information available        |
| <b>Solubility</b>                             | No information available        |
| <b>Partition coefficient; n-octanol/water</b> | No data available               |
| <b>Autoignition Temperature</b>               | No information available        |
| <b>Decomposition Temperature</b>              | No information available        |
| <b>Viscosity</b>                              | Not applicable                  |
| <b>Molecular Formula</b>                      | C7 H8 N2 O2                     |
| <b>Molecular Weight</b>                       | 152.15                          |

**10. Stability and reactivity**

|                                         |                                                                                             |
|-----------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                  |
| <b>Stability</b>                        | Stable under normal conditions.                                                             |
| <b>Conditions to Avoid</b>              | Incompatible products.                                                                      |
| <b>Incompatible Materials</b>           | Acids, Strong oxidizing agents, Acid anhydrides, Acid chlorides, Chloroformates             |
| <b>Hazardous Decomposition Products</b> | Nitrogen oxides (NO <sub>x</sub> ), Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ) |
| <b>Hazardous Polymerization</b>         | No information available.                                                                   |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                               |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component                      | LD50 Oral                | LD50 Dermal | LC50 Inhalation |
|--------------------------------|--------------------------|-------------|-----------------|
| Benzenamine, 2-methyl-5-nitro- | LD50 = 574 mg/kg ( Rat ) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                      | CAS-No  | IARC       | NTP        | ACGIH | OSHA       | Mexico |
|--------------------------------|---------|------------|------------|-------|------------|--------|
| Benzenamine, 2-methyl-5-nitro- | 99-55-8 | Not listed | Not listed | A3    | Not listed | A3     |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** See actual entry in RTECS for complete information.

## 12. Ecological information

### Ecotoxicity

Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

|                                      |                                                                       |
|--------------------------------------|-----------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Insoluble in water                                                    |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                             |
| <b>Mobility</b>                      | Is not likely mobile in the environment due its low water solubility. |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component                                | RCRA - U Series Wastes | RCRA - P Series Wastes |
|------------------------------------------|------------------------|------------------------|
| Benzenamine, 2-methyl-5-nitro- - 99-55-8 | U181                   | -                      |

### 14. Transport information

#### DOT

|                             |                        |
|-----------------------------|------------------------|
| <b>UN-No</b>                | UN2660                 |
| <b>Proper Shipping Name</b> | NITROTOLUIDINES (MONO) |
| <b>Hazard Class</b>         | 6.1                    |
| <b>Packing Group</b>        | III                    |

#### TDG

|                             |                     |
|-----------------------------|---------------------|
| <b>UN-No</b>                | UN2660              |
| <b>Proper Shipping Name</b> | MONONITROTOLUIDINES |
| <b>Hazard Class</b>         | 6.1                 |
| <b>Packing Group</b>        | III                 |

#### IATA

|                             |                        |
|-----------------------------|------------------------|
| <b>UN-No</b>                | UN2660                 |
| <b>Proper Shipping Name</b> | NITROTOLUIDINES (MONO) |
| <b>Hazard Class</b>         | 6.1                    |
| <b>Packing Group</b>        | III                    |

#### IMDG/IMO

|                             |                        |
|-----------------------------|------------------------|
| <b>UN-No</b>                | UN2660                 |
| <b>Proper Shipping Name</b> | NITROTOLUIDINES (MONO) |
| <b>Hazard Class</b>         | 6.1                    |
| <b>Packing Group</b>        | III                    |

### 15. Regulatory information

#### International Inventories

| Component                      | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------------------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Benzenamine, 2-methyl-5-nitro- | 99-55-8 | -   | X    | X    | ACTIVE                                        | 202-765-8 | -      | -   |

| Component                      | CAS-No  | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------------------|---------|-------|----------|------|------|------|------|-------|-------|
| Benzenamine, 2-methyl-5-nitro- | 99-55-8 | X     | KE-24453 | X    | X    | X    | X    | -     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

## Other International Regulations

### Authorisation/Restrictions according to EU REACH

| Component                      | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Benzenamine, 2-methyl-5-nitro- | -                                                                   | Use restricted. See item 43.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |

### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                      | CAS-No  | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------------------|---------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Benzenamine, 2-methyl-5-nitro- | 99-55-8 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                      | CAS-No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Benzenamine, 2-methyl-5-nitro- | 99-55-8 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

### Prepared By

Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

### Revision Date

29-March-2024

### Print Date

29-March-2024

### Revision Summary

New emergency telephone response service provider.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**